

BANK CUSTOMER CHURN ANALYSIS

Executive Dashboard Report

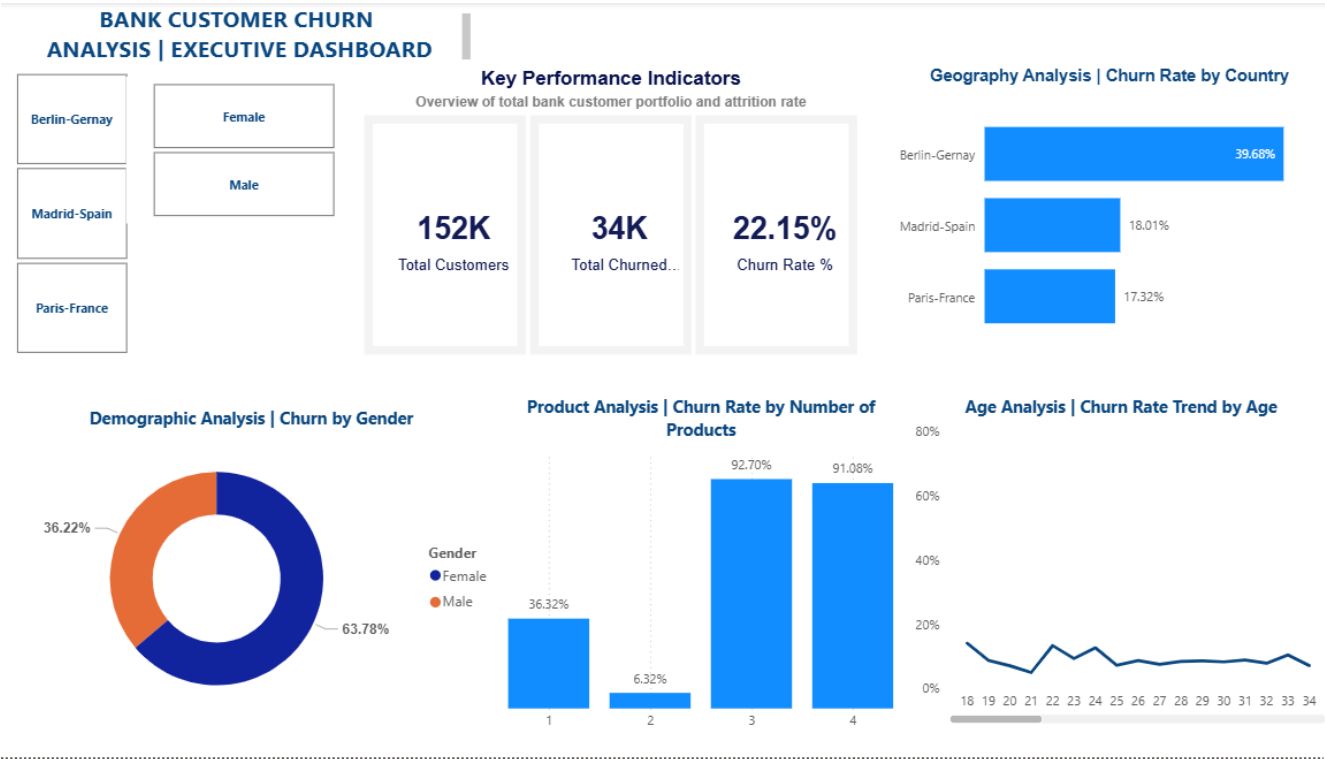


Figure 1: Executive dashboard summarizing total customer base, overall churn rate, and churn breakdown by geography, gender, product holding, and age.

Top 4 Business Insights

Key findings derived from the executive churn dashboard

1. High Attrition in Germany (Geographical Risk)

While churn in France (~17.32%) and Spain (~18.01%) remains under control, Germany (Berlin) shows a critical churn rate of 39.68%. Nearly four out of ten customers in Germany are leaving the bank, pointing to localized service issues or aggressive competitor poaching.

2. Female Customers at Risk (Demographic Attrition)

Of the total churned customer base, 63.78% are female compared to only 36.22% male. The bank's current products and services are disproportionately failing to retain female customers.

3. Multi-Product Customer Dissatisfaction (Product Strategy Gap)

Customers holding exactly two products represent the bank's "sweet spot," with a low churn rate of 6.32%. However, once a customer adopts three or four products, the churn rate spikes sharply to 92.70% and 91.08% respectively, suggesting multi-product holders may be experiencing hidden fees, complex cross-selling, or fragmented service.

4. Strong Youth Retention

Churn remains consistently low (roughly 10-15%) for customers aged 18 to 34, indicating a strong foundation and high brand loyalty among the younger customer base.

Strategic Recommendations

CEO-level actions arising from the insights above

Recommendation 1: Conduct an Operational Audit of the Germany Branch

Action: Launch an immediate, comprehensive review of the German branch's localized products, pricing structures, and customer service protocols. Investigate whether a prominent local competitor is offering superior incentives that are systematically draining the bank's market share.

Recommendation 2: Launch a Women-Centric Customer Retention Program

Action: Run targeted feedback surveys focused on the female demographic to isolate specific pain points. Introduce tailored financial wellness programs, customizable cashback structures, or dedicated relationship managers to re-engage and retain female account holders.

Recommendation 3: Optimize the "2-Product Sweet Spot" and Refine Cross-Selling

Action: Shift marketing away from aggressively pushing three or four products onto a single customer. Standardize high-value two-product combinations (for example, a savings account paired with a credit card). For existing three- and four-product customers, proactively waive maintenance fees or elevate them to a priority service tier to mitigate churn risk.

End-to-End Project Workflow & Approach

Methodology followed from raw data extraction to dashboard delivery

Raw Data Acquisition	Power Query ETL	Data Modeling	DAX Engineering	Interactive Visualization	Business Insight Generation
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1. Data Acquisition

Imported the raw Bank_churn_dataset containing transaction logs, customer demographics, financial tracking parameters, and historical exit indicators.

2. Data Extraction & Transformation (ETL in Power Query)

- Loaded the data into Power Query to isolate multi-variable columns.
- Used custom string delimiters to cleanly separate merged data patterns into individual Geography and Gender columns.
- Cleaned structural text by identifying and replacing legacy syntax markers and stray quotes to normalize raw entries.

3. Advanced Data Cleaning & Profiling

- Replaced text-based "nan" strings with explicit, system-recognized null markers across categorical attributes.
- Applied standard data types to every field, configuring numeric fields such as CreditScore with proper whole-number formats and defined numeric missing values.
- Performed logical range filtering to remove outlier records (e.g., negative or implausible values) in fields like Age, producing a clean 18-100 operational dataset.

4. Data Modeling (Star Schema Preparation)

- Verified table integrity inside Power BI's Model View.
- Maintained a streamlined relational structure and set the Age column to Don't Summarize, ensuring accurate visual cross-filtering and timeline charting.

5. DAX KPI Engineering

- Created a centralized metrics table (_All Measures) to support scalability.
- Built core volume and percentage formulas using COUNT, SUM, and division-safe DIVIDE functions to produce Total Customers, Total Churned Customers, and Churn Rate %.

6. Executive Dashboard Design

- Used a multi-card container to organize high-level KPIs at a glance.
- Structured clean visuals (clustered bar, donut, and trend-line charts) with clear titles, precise decimal formatting, sorted axes, and filters to exclude null values.
- Established a clear visual hierarchy with a prominent dashboard header and cross-filtering slicers for interactive analysis.